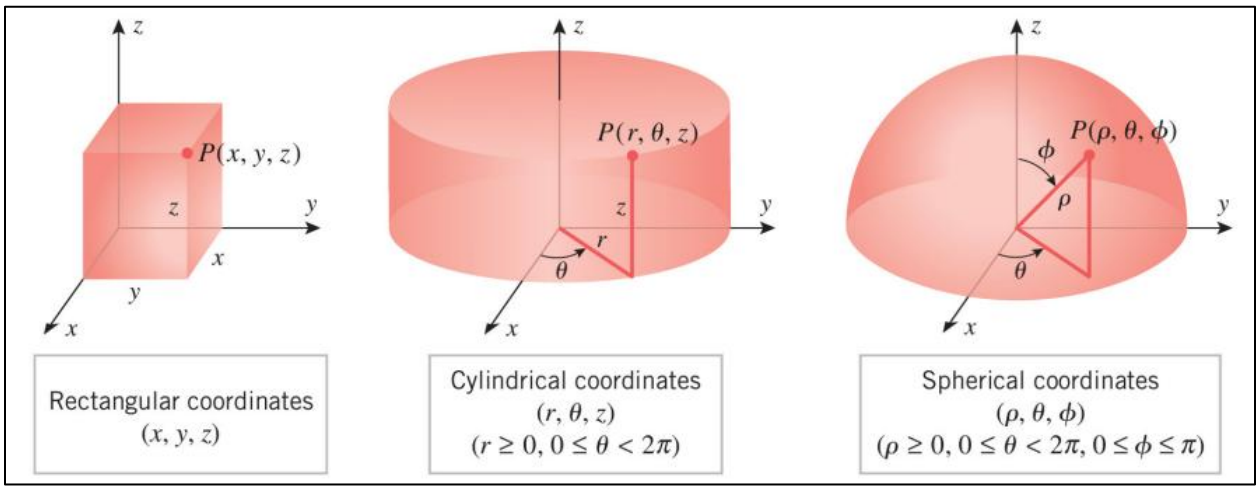
# Lecture 22

## Conversion of Points

Location of a point in 3D-space can be represented in 3 different ways:

1. The rectangular coordinate system
2. The cylindrical coordinate system
3. The spherical coordinate system

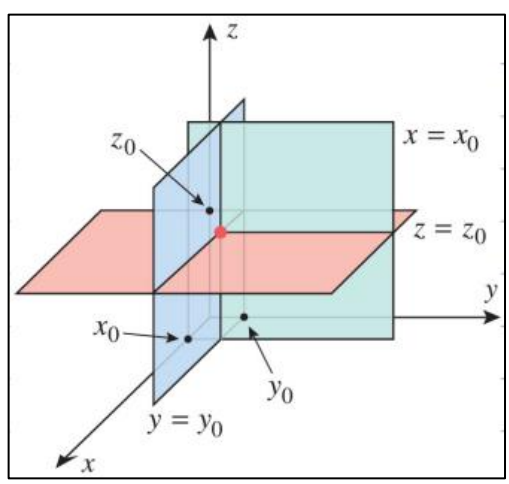



**Constant Surfaces:**

**i) Rectangular Coordinates:**

$x = x_0, \quad y = y_0, \quad z = z_0$

All represent flat planes.

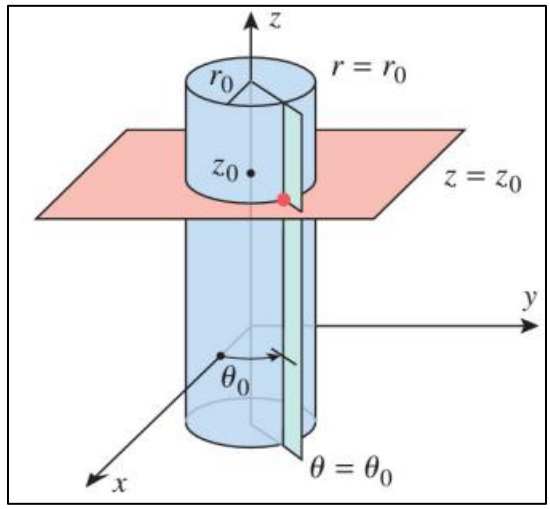


**ii) Cylindrical Coordinates:**

$r = r_0$ represents a cylinder

$\theta = \theta_0$ represents a half plane

$z = z_0$ represents a flat horizontal plane

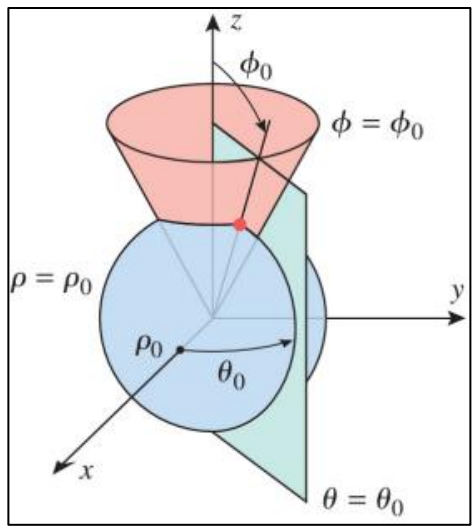



**iii) Spherical Coordinates:**

$\rho = \rho_0$ is a sphere with center $(0,0,0)$ and radius $\rho_0$

$\theta = \theta_0$ represents a half plane

$\phi = \phi_0$ represents a cone




| **Conversion** | **Formulas** |
|---|---|
| **Cylindrical to Rectangular:** $(r,\theta,z) \to (x,y,z)$ | $x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$ |
| **Rectangular to Cylindrical:** $(x,y,z) \to (r,\theta,z)$ | $r = \sqrt{x^2+y^2}, \quad \tan\theta = \frac{y}{x}, \quad z = z$ |
| **Spherical to Cylindrical:** $(\rho,\theta,\phi) \to (r,\theta,z)$ | $r = \rho\sin\phi, \quad \theta = \theta, \quad z = \rho\cos\phi$ |
| **Cylindrical to Spherical:** $(r,\theta,z) \to (\rho,\theta,\phi)$ | $\rho = \sqrt{r^2+z^2}, \quad \theta = \theta, \quad \tan\phi = \frac{r}{z}$ |
| **Spherical to Rectangular:** $(\rho,\theta,\phi) \to (x,y,z)$ | $x = \rho\sin\phi\cos\theta, \quad y = \rho\sin\phi\sin\theta, \quad z = \rho\cos\phi$ |
| **Rectangular to Spherical:** $(x,y,z) \to (\rho,\theta,\phi)$ | $\rho = \sqrt{x^2+y^2+z^2}, \quad \tan\theta = \frac{y}{x},$ $\cos\phi = \frac{z}{\sqrt{x^2+y^2+z^2}}$ |

**Value of $\theta$ in Different Quadrants:**

- First quadrant: $\tan^{-1}\left(\left|\frac{y}{x}\right|\right)$
- Second quadrant: $\pi - \tan^{-1}\left(\left|\frac{y}{x}\right|\right)$
- Third quadrant: $-\pi + \tan^{-1}\left(\left|\frac{y}{x}\right|\right)$
- Fourth quadrant: $-\tan^{-1}\left(\left|\frac{y}{x}\right|\right)$

**Example:** Convert from Rectangular to Spherical: $\left(1, -\sqrt{3}, -2\right)$

**Solution:** Given, $(x, y, z) = \left(1, -\sqrt{3}, -2\right)$

So, $x = 1, y = -\sqrt{3}, z = -2$.

We know that,

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\therefore \rho = \sqrt{1^2 + \left(-\sqrt{3}\right)^2 + (-2)^2} = \sqrt{1 + 3 + 4} = \sqrt{8} = 2\sqrt{2}$$

Now,

$$\tan\theta = \frac{y}{x}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$\Rightarrow \theta = -\tan^{-1}\left(\left|\frac{-\sqrt{3}}{1}\right|\right)$$

$$\therefore \theta = -\frac{\pi}{3}$$

Then,

$$\cos\phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\Rightarrow \cos\phi = \frac{-2}{\sqrt{1^2 + \left(-\sqrt{3}\right)^2 + (-2)^2}}$$

$$\Rightarrow \cos\phi = \frac{-2}{\sqrt{1 + 3 + 4}}$$

$$\Rightarrow \cos\phi = \frac{-2}{\sqrt{8}}$$

$$\Rightarrow \cos\phi = \frac{-2}{2\sqrt{2}}$$

$$\Rightarrow \cos\phi = \frac{-1}{\sqrt{2}}$$

$$\Rightarrow \phi = \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right)$$

$$\therefore \phi = \frac{3\pi}{4}$$

$$\therefore (\rho, \theta, \phi) = \left(2\sqrt{2}, -\frac{\pi}{3}, \frac{3\pi}{4}\right)$$

**Example:** Convert from Cylindrical to Spherical: $\left(4, \frac{5\pi}{6}, 4\right)$

**Solution:** Given, $(r, \theta, z) = \left(4, \frac{5\pi}{6}, 4\right)$

$\therefore r = 4, \theta = \frac{5\pi}{6}, z = 4.$

We know that,

$$\rho = \sqrt{r^2 + z^2} = \sqrt{4^2 + 4^2} = \sqrt{16 + 16} = \sqrt{32} = 4\sqrt{2}$$

$$\theta = \theta = \frac{5\pi}{6}$$

$$\tan\phi = \frac{r}{z} = \frac{4}{4} = 1$$

$$\Rightarrow \phi = \tan^{-1}(1) = \frac{\pi}{4}$$

$$\therefore (\rho, \theta, \phi) = \left(4\sqrt{2}, \frac{5\pi}{6}, \frac{\pi}{4}\right)$$

**Example:** Convert from Rectangular to Cylindrical: $\left(\sqrt{2}, -\sqrt{2}, -1\right)$

**Solution:** Given, $(x, y, z) = \left(\sqrt{2}, -\sqrt{2}, -1\right)$

$\therefore x = \sqrt{2}, y = -\sqrt{2}, z = -1.$

We know that,

$$r = \sqrt{x^2 + y^2} = \sqrt{\left(\sqrt{2}\right)^2 + \left(-\sqrt{2}\right)^2} = \sqrt{2 + 2} = \sqrt{4} = 2$$

$\tan\theta = \frac{y}{x}$

$\Rightarrow \theta = -\tan^{-1}\left(\left|\frac{-\sqrt{2}}{\sqrt{2}}\right|\right)$

$\Rightarrow \theta = -\tan^{-1}(1)$

$\therefore \theta = -\frac{\pi}{4}$

$z = z = 1$

$\therefore (r, \theta, z) = \left(2, -\frac{\pi}{4}, 1\right)$

**Example:** Convert from Spherical to Rectangular: $\left(1, \frac{2\pi}{3}, \frac{3\pi}{4}\right)$

**Solution:** Given, $(\rho, \theta, \phi) = \left(1, \frac{2\pi}{3}, \frac{3\pi}{4}\right)$

$\therefore \rho = 1, \theta = \frac{2\pi}{3}, \phi = \frac{3\pi}{4}$

We know that,

$x = \rho\sin\phi\cos\theta = 1 \times \sin\frac{3\pi}{4} \times \cos\frac{2\pi}{3} = 1 \times \frac{1}{\sqrt{2}} \times -\frac{1}{2} = -\frac{1}{2\sqrt{2}}$

$y = \rho\sin\phi\sin\theta = 1 \times \sin\frac{3\pi}{4} \times \sin\frac{2\pi}{3} = 1 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{4}$

$z = \rho\cos\phi = 1 \times \cos\frac{3\pi}{4} = 1 \times \frac{-1}{\sqrt{2}} = -\frac{1}{\sqrt{2}}$

$\therefore (x, y, z) = \left(-\frac{1}{2\sqrt{2}}, \frac{\sqrt{6}}{4}, -\frac{1}{\sqrt{2}}\right)$

## H.W. Exercise 11.8 (1-12)